CausalCache: Conditional High-Fidelity Restoration for Long-Horizon GUI Agents

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Abstract

Autonomous GUI agents must carry task-critical visual information across long workflows, yet can activate only a few past screenshots. A complete interaction trace can be retained cheaply as textual summaries, creating a distinct memory problem: not which events remain recorded, but which summarized events should be restored to high-fidelity pixels. We formalize this problem as *conditional fidelity restoration* under a fixed image budget. Recent-B assigns every high-fidelity slot to the latest events. CausalCache starts from this recent allocation and replaces a slot only when a distant event is predicted to be more useful than the recent event it displaces. Its history-gated key/value adapter (HGKV) changes only restored history-image tokens and exactly bypasses when no history image is active; a budget-aware selector then reallocates the same high-fidelity slots over the complete summarized trace. Matched-budget relevant, recent, and wrong restorations train HGKV to use history content selectively rather than amplify restored pixels indiscriminately. Our evaluation separates three questions: whether high-fidelity history is useful, whether the policy can consume restored pixels selectively, and whether conditional reallocation improves over recency at the same budget. On OSWorld-Verified, high-fidelity history improves success by about 13 points over summary-only memory, while same-budget allocations remain indistinguishable. Zero-shot on MobileWorld's construction-defined memory-critical split, CausalCache improves over Recent-B by 8.6 points, remains neutral on matched controls, and yields a significant method-by-memory-criticality interaction.

Introduction

Autonomous GUI agents that interpret natural-language instructions and operate computing devices could provide substantial value by automating repetitive work, augmenting human capabilities, and completing complex workflows across applications. Realizing this promise requires agents to act over long horizons: a single instruction may require navigating several interfaces, collecting values from earlier screens, and using them many steps later.

Long-horizon interaction creates a visual-context bottleneck. At every step, a GUI agent accumulates screenshots, actions, text arguments, coordinates, and interface changes. The complete interaction trace can be preserved cheaply as compact textual summaries, but keeping every screenshot active consumes substantial multimodal context and can distract the policy. A past event can therefore remain available at two policy-visible fidelities: a low-fidelity summary, or the same summary paired with its archived screenshot.

This setting exposes a memory question that is easy to conflate with retention or retrieval: once every event is already preserved in the summary trace, which events should be shown again in pixels? We call this *conditional high-fidelity restoration*. Restoration reattaches archived screenshots; it does not reconstruct pixels from text. The active image budget is fixed, so restoring a distant event requires demoting a currently visible event back to summary-only form. Recent-B, the natural baseline, assigns all high-fidelity slots to the latest events. CausalCache treats recency as the fallback rather than the objective: it replaces a recent slot only when a distant event is predicted to be more useful in the context of the images already active.

Figure 1 illustrates the distinction. The task moves from an order application to a messaging application and later requires the product names and order number shown on an earlier screen. Both policies keep the complete summary trace and expose exactly four history images. Recent-4 uses all four slots for the latest events and loses the task-critical order screen; CausalCache restores that earlier event by giving up less useful recent images, then completes the message correctly. The comparison adds no visual capacity—only the identities of the four high-fidelity events change.

Choosing better events is only half of the problem. A restoration selector cannot improve behavior if the action policy does not use the content of the reattached screenshots. In controlled replacements, we find that the frozen policy responds to the presence of historical images but does not reliably prefer a task-relevant distant screenshot over the informative recent frame it would displace. The central claim of this paper is therefore conditional: a distant event should be restored only when it is more useful than the recent event it replaces, and the policy must be able to consume the restored pixels selectively.

CausalCache addresses these two requirements with two learned components. First, a history-gated key/value interface (HGKV) applies low-rank updates only to restored

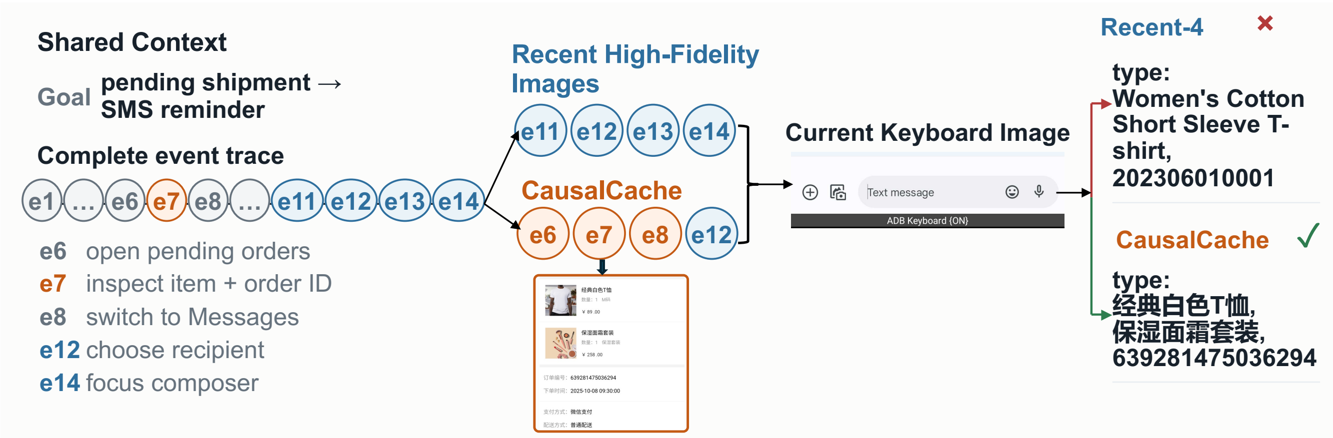

Figure 1: A real cross-app `CartInfoNotificationTask` episode. The agent must inspect items awaiting shipment in TaoDian, then send an SMS containing only the product names and order number. Both rollouts retain the complete summary trace and use four active history-image slots. Recent-4 exposes $[e11, e12, e13, e14]$. CausalCache-P instead exposes $[e6, e7, e8, e12]$, restoring the enlarged $e7$ order screenshot before the shared message-composition stage. Recent-4 types the wrong item and order number and fails; CausalCache types the visually recovered values and succeeds. This episode illustrates fixed-budget fidelity reallocation rather than an aggregate effect.

history-image tokens; instruction tokens, event summaries, the current screen, and action tokens bypass the adapter, and the no-history path is exactly the frozen policy. We train this interface on matched-budget recent, relevant, and wrong restorations, anchoring every arm to its own frozen score so that uniformly amplifying historical pixels earns no credit. Second, a budget-aware selector scores summarized events conditioned on the current allocation and starts from Recent-B. A distant event is promoted only by outscoring the recent visual realization it would replace. The default online variant, CausalCache-P, uses the policy's tentative action as a reference and invokes a second decision only when the allocation changes.

Our experiments are organized around three claims. *First*, high-fidelity history *availability* can be valuable, but this does not imply that recent screenshots are the best allocation. *Second*, HGKV learns content-selective use of restored pixels while preserving the base policy on no-history, recent-history, and wrong-history inputs. *Third*, conditional *allocation* improves over Recent-B at the same image budget when success depends on distant visual evidence, while remaining neutral on matched controls without that dependency. This decomposition keeps "more visual memory," "using visual memory," and "choosing the right visual memory" as separate empirical questions.

Our contributions are:

- We formulate long-horizon GUI memory as fixed-budget conditional fidelity restoration: every event remains in a complete summary trace, while the controller decides which summarized events regain their archived screenshots.
- We introduce a two-component solution consisting of a policy-preserving HGKV interface for selectively consuming restored pixels and a budget-aware selector for reallocating high-fidelity slots over the complete trace.
- We provide matched-budget and matched-control evidence that separates high-fidelity availability, policy-side consumption, and conditional allocation, including zero-shot closed-loop evaluation on desktop and mobile benchmarks.

Related Work

Visual GUI agents and benchmarks. WebShop, WebArena, Mind2Web, WorkArena, WorkArena++, VisualWebArena, and REAL span grounded shopping, real websites, enterprise workflows, and visual web interaction (Yao et al. 2022; Zhou et al. 2024; Deng et al. 2023; Drouin et al. 2024; Boisvert et al. 2024; Koh et al. 2024; Garg et al. 2025). AITW, AndroidControl, AndroidWorld, OSWorld, and GUI-Odyssey extend executable evaluation to mobile and desktop interfaces (Rawles et al. 2023; Li et al. 2024a; Rawles et al. 2025; Xie et al. 2024; Lu et al. 2025). Screenshot-native agents learn grounding and actions directly from pixels (Hong et al. 2024; Cheng et al. 2024; Lin et al. 2025; Xu et al. 2026). These works measure long-horizon interaction; our question is which completed event should re-enter a bounded prompt in pixels. MobileWorld further stresses long-horizon, cross-application mobile workflows under reproducible functional evaluation (Kong et al. 2026).

Agent memory and reusable experience. RAG retrieves external memory, ReAct interleaves reasoning with action, and later agents preserve reflections, experience, or evolving long-term stores (Lewis et al. 2020; Yao et al. 2023; Shinn et al. 2023; Zhao et al. 2024; Zhong et al. 2024; Gutiérrez et al. 2024). ICAL retrieves multimodal programs of thought (Sarch et al. 2024). These systems write or retrieve artifacts across interactions or tasks; CausalCache instead holds a within-episode summary trace fixed and controls whether an existing event is exposed as summary only or summary plus image.

Long-horizon GUI memory. GUI-Odyssey resamples prior screenshots, MementoGUI learns memory operators, and AndroTMem retrieves causally linked anchor states (Lu et al. 2025; Zeng et al. 2026; Shi et al. 2026). Their primary variable is which item is retained or retrieved. CausalCache keeps the compressed trace fixed, controls event fidelity conditioned on the already-active visual context, and trains the interface that consumes restored pixels.

Context and visual-token compression. Prompt compressors remove textual redundancy (Jiang et al. 2023, 2024; Pan et al. 2024); KV methods retain, evict, select, or quantize cache states (Xiao et al. 2024; Zhang et al. 2023; Liu et al. 2023; Li et al. 2024b; Liu et al. 2024; Corallo and Papotti 2024); and visual methods merge or prune tokens within an image (Bolya et al. 2023; Chen et al. 2024; Yang et al. 2025). Their unit is a token or KV position. CausalCache instead preserves the summary trace and allocates whole archived screenshots across events; HGKV controls only how promoted history-image tokens are consumed.

Interventional memory value. Recent work studies memory through controlled interventions or active state influence (Srivastava 2026; Liu et al. 2026). Our intervention is narrower and executable: a selected event regains one archived post-action image while the remaining prompt stays fixed, yielding a policy-specific behavioral surrogate rather than an environment-level causal effect. Set models parameterize coalition-conditioned scores (Zaheer et al. 2017; Lee et al. 2019), whose marginals connect to cooperative-game attribution (Shapley 1953); we use controlled restorations to train a deployable selector for one fixed GUI policy.

Method

Problem Formulation

At decision t, the agent receives instruction g, current observation x_t, and history $H_t = \{e_1, \dots, e_{t-1}\}$ with $e_j = (x_j, a_j, x_{j+1})$, where each event has a compact low-fidelity representation $e_j^{\text{low}} = \tilde{e}_j$, its one-line action summary, and an available high-fidelity representation $e_j^{\text{high}} = (r_j, v_j)$: the official retained-turn form pairing the step's verbatim policy response r_j with its reattached archived post-action screenshot v_j. The official protocol pairs every retained observation with the response that produced it, so promotion and demotion move an event between these two forms as a unit — a fidelity allocation reassigns retained turns, not images alone.

Fixed active-fidelity budget. Every prompt contains the goal g, the *complete* textual action summaries $\{\tilde{e}_1, \dots, \tilde{e}_{t-1}\}$, and the current screenshot x_t. An active visual-context budget B limits how many summarized events may be promoted by reattaching their archived pixels; it is not a limit on cold archival storage. A restoration allocation is a set $S \subseteq \{1, \dots, t-1\}$ with $|S| \le B$. Let $Q(S)$ denote the policy's mean target-token log-likelihood of the successful next action when events in S are exposed in summary-plus-image form. The standard baseline is $S_{\text{recent}} = \text{Recent-}B$, the B most recent distinct events promoted to high fidelity. CausalCache targets

$$S^* = \arg \max_{\substack{S \subseteq \{1,\dots,t-1\} \\ |S| \le B}} Q(S), \qquad Q(S^*) > Q(S_{\text{recent}}), \tag{1}$$

so recency is one candidate fidelity allocation, not a separate modality. The realized replacement intensity $k = |S \setminus \text{Recent-}B|$ is a measured statistic, not a preset parameter. All comparisons at a given B hold the image count, resolutions, underlying event trace, and prompt structure fixed: every allocation renders exactly $|S|$ retained turns and keeps every other event at summary fidelity; only which events occupy the high-fidelity form differs. Cross-B comparisons are analysis, not the causal contrast. Given a partial allocation S, conditional marginals $\Delta_t(j \mid S) = Q(S \cup \{j\}) - Q(S)$ drive selection, and a replacement is justified only when the distant event's restoration marginal exceeds that of the recent visual realization it displaces. Every formally reported set utility reruns the complete restored set; we never report a sum of singleton gains as the utility of a coalition.

Policy-Preserving History-Gated KV Adapter

Let h_l be the input to layer l's attention projections. In the last eight language-model layers, CausalCache adds rank-8, α=16 low-rank residuals (Hu et al. 2022) only to key and value projections:

$$\begin{aligned} K_l &= W_l^K h_l + M_{\text{hist}} \, \Delta W_l^K h_l, \\ V_l &= W_l^V h_l + M_{\text{hist}} \, \Delta W_l^V h_l. \end{aligned} \tag{2}$$

M_{hist} is a token mask that is true only for image tokens reattached to promoted historical events. It excludes instruction tokens, event-summary text, the current observation image, and action tokens. The mask is constructed fail-closed from the multimodal token-type sequence and per-image patch geometry; any mismatch between declared images and encoded image blocks aborts the forward pass. When no historical event is promoted, the adapter context is absent and the projection hook returns the original module output, so $\pi_{\text{HG}}(\cdot \mid S{=}\varnothing) = \pi_0(\cdot \mid S{=}\varnothing)$ up to bitwise equality in our implementation.

Per-B Fidelity-Restoration Supervision

Training states are decision points of successful desktop trajectories in which the target action a_t^* recurs earlier in the same trajectory under full-action equivalence (type, arguments, and coordinates within tolerance); the screenshot preceding the earlier occurrence is the target-specific restoration candidate. For each state and each budget B we construct three matched-budget fidelity allocations with identical summaries, current screen, and target—every prompt carries exactly B history images:

- **Recent**: Recent-B, the B most recent distinct events promoted to summary-plus-image form;
- **Relevant restore**: Recent-$(B{-}1)$ + $\{v_{j+}\}$, the oldest recent visual slot replaced by the target-specific archived screenshot (age $\ge$ B+2, strictly older than the whole window);

- **Wrong restore**: the same slot replaced by an age-matched same-trajectory screenshot whose following action is *not* equivalent to a_t^*.

Positives do not require the base action to be wrong: both information addition and evidence amplification are admissible mechanisms.

Why single-slot replacement (k=1). Training teaches only the k=1 replacement interface, and this is an empirically grounded interface choice rather than a structural limit. On the demand side, multi-frame needs are rare even where archived pixels matter most: under the strictest *image-only* criterion—the required value appears in neither the goal, the current frame, the recent window, nor any completed step's action text—mining 1,326 GUI-Odyssey trajectories yields 96 image-only decision points at B=4, of which only two require two distinct evidence frames simultaneously (3 of 239 at B=1); on the desktop corpus, 75% of eligible states admit exactly one recurrence witness. On the utility side, a matched-budget probe under the official multi-turn protocol shows the first replacement is worth its slot, while the paired increments of a second and third replacement are indistinguishable from zero or negative—under a fixed budget each further distant promotion must evict the next-most-recent visual realization, and its measured value no longer covers that price. Larger-k allocations nonetheless remain expressible at deployment: the selector composes exactly-B allocations with exact re-scoring, defaulting unclaimed slots to recent events, so k stays a result statistic. Supervision is annotation-free, so a domain that does exhibit multi-frame demand (detectable by the same probe) requires only regenerating the corpus with $k \geq 2$ restoration arms: a data-mix change, not a method change.

Let ℓ denote mean target log-likelihood, superscripted by adapter state (active or frozen bypass). Each arm is anchored to *its own* frozen score:

$$A_s = \ell^{\text{on}}_{\text{relevant}} - \ell^{\text{off}}_{\text{relevant}}, \qquad A_r = \ell^{\text{on}}_{\text{recent}} - \ell^{\text{off}}_{\text{recent}}, \qquad A_n = \ell^{\text{on}}_{\text{wrong}} - \ell^{\text{off}}_{\text{wrong}}, \tag{3}$$

and the DiD objective is

$$\begin{aligned}\mathcal{L} = {} & [m - (A_s - A_r)]_+ + [m - A_s]_+ \\ & + [m - (A_s - A_n)]_+ \\ & + \lambda_{\text{cap}}\big([|A_r| - \epsilon]_+ + [|A_n| - \epsilon]_+\big) + \lambda \|\Delta W\|_2^2,\end{aligned} \tag{4}$$

with m=0.01, ϵ=0.02, λ_{cap}=2, λ=10^{-4}, and no action cross-entropy. On the identity adapter every increment is zero, so $A_s - A_r$ is exactly zero: the frozen policy's own preference for target-relevant restorations cannot masquerade as adapter skill, and uniformly amplifying all restored pixels buys nothing. The dead-zone caps bound recent-window and wrong-history drift with gradients on the same scale as the selection hinges. Losses and evaluation are stratified by B; a pooled mean could hide a method that helps at B=4 but fails at B=1.

Success-Anchored Utility and Selector Supervision

After the history interface is frozen, selector labels are true policy utilities: for each state, singleton utilities for the *entire* summarized event pool (recent events included)—an exhaustive B=1 oracle—the matched $Q(\text{Recent-}B)$ anchors, and, for larger budgets, set utilities along beam-3 teacher paths over a singleton-shortlisted pool. Every set is fully re-scored as a complete prompt through the frozen HGKV policy; set utility is never approximated by summing singleton utilities.

Budget-Aware Restoration Selector

The selector is a single set-conditional marginal scorer $\widehat{\Delta}_t(j \mid S)$ trained on the true marginals implied by the label table (singleton edges give S=$\varnothing$; each beam edge gives its parent set), with a joint regression and within-state ranking loss. Its input encodes, explicitly, candidate age and recency-window membership, target-recurrence witness signals with coordinate distance, action family, frame-deduplication multiplicity, trajectory shape, and set context (selected-set size, overlap and age gaps between the candidate and the already-selected events).

Deployment composes exactly-B fidelity allocations by beam search initialized at $\text{Recent-}B$: every summarized event is a candidate, and a distant event is promoted only by displacing the recent visual realization whose predicted marginal it beats, realizing Eq. 1 as a relative comparison with no learned stopping threshold—declining every replacement recovers Recent-B exactly. Selector labels cover $B \in \{1, 2, 4\}$; the realized replacement intensity k is reported per budget, never preset.

Reference actions and the two deployment modes. Four of the selector's input features are *witness* features: they test whether the action taken right after some archived event matches a reference action. At training time the reference is the gold teacher-forced target—an offline oracle that upper-bounds what any online reference can provide but does not exist at deployment. We therefore instantiate two online modes that share every trained weight and differ only in the reference. **CausalCache-P** (default) uses the current policy's tentative action: the policy first acts on the Recent-B prompt, the tentative action serves as a platform-calibrated reference, and the policy re-decides on the re-allocated prompt when a recent visual realization is displaced. **CausalCache-LA** replaces the proposal with *the action just executed* at step $t-1$, eliminating the second policy call: memory is selected before the single generation, adding only the selector's own cost ($<$ 10 ms). On held-out desktop decision groups the two modes are statistically indistinguishable from each other and from the gold reference, but LA fails to transfer to the unseen mobile benchmark, where P attains the higher mean. We therefore treat LA as an efficiency variant rather than the primary method; the directional result is consistent with the reference action being one bottleneck for cross-platform transfer.

A ladder of weaker references isolates what the action signal buys. *No reference* (witness features zeroed) keeps B=2 significant but loses B=4 while roughly tripling full-replacement allocations—without any action signal the selector cannot tell which summarized event merits visual restoration and over-replaces. *Instruction-only intent* (frozen-embedding similarities between the instruction, each candidate's action line, and recent behavior) repairs the over-replacement pathology yet stays non-significant at $B{=}4$. Action-conditioned references thus dominate instruction-only and reference-free variants; their internal order depends on platform and protocol, which is why the deployed default re-grounds the reference in the policy's own proposal.

Experiments

Experimental Setup

Training Data and Model Selection. Policy adaptation and selector training use only desktop data: successful AgentNet/OpenCUA Ubuntu trajectories (5,000 screened, 2,293 successful, 6,003 decision points) yield fixed-budget replacement groups at $B{=}1/2/4$ of $969/764/470$ (2,203 in total; 1,774 training units and 211 development groups, trajectory-disjoint; click 52–56%, plus key/type/drag/scroll), complemented by a high-precision OSWorld witness seed mined from official successful trajectories. A further 160 groups at $B{=}8$ are constructed but deliberately excluded from training, so $B{=}8$ evaluates out-of-training budget extrapolation. Checkpoints and thresholds are selected only on the desktop development split under pre-registered gates: positive DiD selection with a positive bootstrap lower bound, and drift caps $|A_r|, |A_n| < \epsilon$. No mobile benchmark data touches training or selection. All adapter training, offline scoring, and closed-loop serving run on NVIDIA H200 Tensor Core GPUs.

The Fixed-Budget Fidelity-Restoration Design. Figure 2 shows the two learned components and the same-budget re-decision that defines the core comparison. Every compared allocation carries exactly B history images with identical resolutions, summaries, and prompt structure, so the contrast isolates *which summarized events are promoted to high fidelity*. $B{=}4$ is the primary deployment-like setting; the realized number of non-recent promotions k is measured rather than preset.

Policies and Fidelity-Allocation Baselines. The frozen policy is GUI-Owl-1.5-8B-Instruct (Xu et al. 2026). Under identical data, steps, and cadence we compare the frozen policy, an ungated KV control (the same last-eight layers, K/V targets, rank, and alpha as HGKV but active on all tokens), and HGKV itself (token-gated, structurally bypassed at $B{=}0$). Fidelity-allocation baselines are Recent-B, Random, OCR/RGB similarity, and the learned budget-aware restoration selector (exact-B, recent fallback); a beam oracle over the candidate pool bounds headroom. Every method receives the same summaries and the same active image budget.

Zero-Shot Benchmarks and Metrics. The policy interface and selector, trained on desktop data only, are evaluated (i) on held-out desktop decision groups (offline, per-B), (ii) zero-shot on an uncontaminated OSWorld task roster (Xie et al. 2024) with official evaluator scores (tasks whose trajectories seeded the witness corpus are excluded from zero-shot claims and reported separately as in-domain diagnostics), and (iii) zero-shot on a cross-app mobile memory benchmark of 117 runnable tasks (62 cross-app memory candidates, 55 single-app controls), which contributes no data to any training or selection decision (Kong et al. 2026). Offline metrics are teacher-forced log-likelihood margins and full-action equivalence under an argument and coordinate tolerance (sensitivity settings in the supplement); closed-loop metrics are paired task success with cluster bootstrap, step counts, and wall time.

Results

Claim 1: High-fidelity availability and fidelity allocation are distinct. The closed-loop desktop result first establishes *availability*. On OSWorld-Verified, every high-fidelity arm reaches 32.7–34.1%, improving by roughly 13–14 points over summary-only memory (19.9%), with paired confidence intervals excluding zero. However, the same-budget high-fidelity allocations are statistically indistinguishable. OSWorld therefore shows that historical pixels can matter, but it does not by itself show that conditional reallocation is better than recency.

Controlled desktop replacements expose the missing ingredient. Against a redundant recent image, the frozen policy prefers a target-relevant archived screenshot; against the informative true previous frame it would actually displace, that preference vanishes. The frozen replacement effect further decays as the recent window grows and reverses at the untrained largest budget (Table 1). Thus, selecting a plausible distant event is not enough: the action policy does not inherit the content selectivity needed to profitably trade that event against a useful recent frame.

Claim 2: HGKV learns selective history use without broad policy drift. Table 1 reports the pre-registered desktop results. HGKV learns a positive difference-in-differences selection effect: activating the adapter helps the relevant restoration more than the matched recent and wrong restorations. At the selected checkpoint, recent-history and wrong-history drift remain inside the pre-specified cap, while the no-history path is bitwise identical to the frozen policy. The layer- and rank-matched ungated KV control also learns some selectivity, showing that the adapted K/V projections carry useful signal, but token gating yields the larger paired selection effect while keeping unrelated tokens outside the adaptation path. Across the trained budgets, the gated interface maintains positive policy-side selectivity under the fixed-budget protocol (Table 1); full per-budget intervals are reported in the supplement.

Claim 3: Reallocation helps when success depends on distant visual evidence. Offline, the learned selector improves over Recent-B under fresh full-set policy scoring, whereas random exact-budget restoration and RGB similarity remain in the noise band (Table 3). This rules out the weaker explanations that any archived image helps or that visual similarity is a sufficient allocation rule.

The primary closed-loop allocation test is MobileWorld's construction-defined split. The benchmark contributes no training or model selection signal and contains 62 cross-application memory-critical tasks, where required information appears in an earlier application and has left the recent window, plus 55 matched single-application controls without that dependency. At the primary budget, CausalCache-

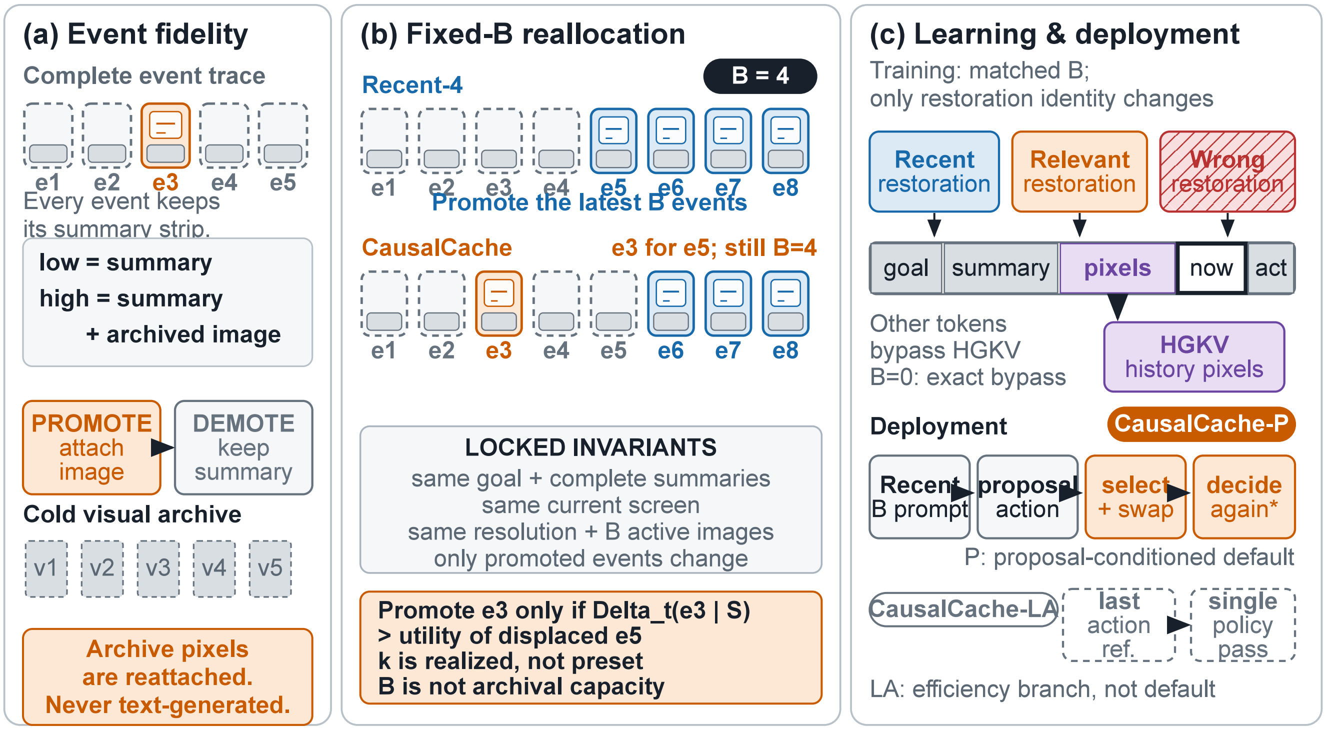

Figure 2: Overview of conditional high-fidelity restoration. (a) Every event remains in the complete trace as a textual summary and is linked to an archived screenshot; promotion changes event fidelity rather than event inclusion. (b) Recent-B spends the fixed image budget on the latest events, whereas CausalCache reallocates the same slots and restores a distant event only when it is predicted to be more useful than the recent event it displaces. The goal, complete summaries, current screen, image count, and image resolutions are held fixed. (c) Matched-budget recent, relevant, and wrong restorations train HGKV to modify only restored history-image tokens. At deployment, CausalCache-P begins from Recent-B, uses a tentative action to select and swap events, and re-decides only if the allocation changes. CausalCache-LA is a single-pass efficiency variant rather than the default.

Fixed-budget fidelity allocation (offline desktop heldout; primary setting $B = 4$)

Policy	Allocation	B=1	B=2	B=4	B=8[†]	Realized k	Notes
Frozen	Recent-B	0	0	0	0	0	reference
Frozen	oldest slot → archived	+.079	+.026	−.000	−.031	1	
HGKV	Recent-B	+.002	+.005	+.007	+.009	0	drift-capped
HGKV	oldest slot → archived	+.103	+.047	+.018	−.016	1	gate passed
HGKV	selector, LA reference (deployed single-pass)	+.039	+.025	+.015	−.005	measured	71% displace ≥1 at B=4
HGKV	selector, gold reference (offline upper bd.)	+.044	+.025	+.011	−.002	measured	teacher-forced oracle

Table 1: Fixed-budget fidelity-allocation results. Each cell is a matched-count contrast on the desktop heldout split; every allocation promotes exactly B events to summary-plus-image form, and the learned selector may keep every promotion on the recent window (k=0); [†]B=8 is excluded from training and reports budget extrapolation. Cross-B comparisons are analysis only.

P improves over the same-budget Recent-B allocation from 19.4% to 28.0% on the memory-critical tasks, a gain of 8.6 points. On the controls it remains neutral (40.6% versus 42.4%), and the method-by-memory-criticality interaction is significant (two-sided p=0.022). On the full 117-task roster, CausalCache-P attains 33.9%, compared with 30.2% for HGKV+Recent-B and 28.2% for the frozen summary-only policy (Table 2). Frozen Recent-B, HGKV+Recent-B, and summary-only memory are statistically indistinguishable, so the mobile gain is carried by which events are promoted, not by adding the adapter to a recent allocation.

Budget robustness and deployment cost. **[Pending: Insert the final paired MobileWorld budget sweep at $B \in \{1, 2, 8\}$, including completed run counts, confidence intervals, and the method-by-budget analysis. Partial cam-**

Method	OSWorld-Verified (361)	MobileWorld (117)	Extra policy calls
Frozen, summary-only (B=0)	19.9	28.2 (3 runs)	0
Frozen + Recent-B	32.7	29.9	0
HGKV + Recent-B	33.0	30.2 (3 runs)	0
CausalCache-LA (single-pass)	**34.1**	29.1	0
CausalCache-P (default)	32.7	**33.9** (3 runs)	conditional

Table 2: Closed-loop success rates (%) at the primary deployment budget B=4; the budget axis is swept offline in Table 1. OSWorld-Verified: official no-GDrive roster and official 15-step budget, one run per arm. MobileWorld: zero-shot three-run means. High-fidelity arms beat summary-only memory on desktop but remain pairwise indistinguishable; CausalCache-P is best on mobile. Full paired statistics and per-round results are in the supplementary document.

Allocation rule	B=2	B=4
Learned scorer	$+.030$ $[+.019, +.041]$	$+.017$ $[+.007, +.028]$
Random exact-B	$+.006$ $[-.006, +.017]$	$+.008$ $[-.001, +.018]$
RGB similarity	$+.006$ $[-.003, +.015]$	$+.007$ $[-.001, +.015]$
Inverted score	$-.016$ $[-.029, -.003]$	$-.009$ $[-.018, -.000]$
No recency feats	$+.012$ $[-.000, +.025]$	$+.008$ $[-.001, +.017]$

Table 3: Chooser ladder: log-likelihood margin over Recent-B at matched budgets with fresh full-set policy scoring (95% episode-clustered CIs). Only the learned conditional ranking clears the noise band; reversing it is symmetrically harmful.

B	Recent-B	CausalCache-P
0 (summary-only)	28.2	—
1	TBD	TBD
2	TBD	TBD
4 (primary)	30.2	33.9
8 (untrained)	TBD	TBD

Table 4: Closed-loop MobileWorld success (%) along the budget axis. The completed primary setting is reported at B=4; unfinished budgets are shown only as placeholders and will be replaced by final multi-run estimates with paired confidence intervals.

paigns should not be reported as point estimates.] Across 2,224 audited mobile steps at the completed primary setting, the conditional second pass fires on 82% of steps, and the selector itself costs under 40 ms against a 4.2 s median policy pass. The re-decision adds 66% to policy compute per affected step; because environment interaction dominates end-to-end execution, the measured median task-level overhead is 7% on desktop. Complete intervals and the witness-reference ladder are in the supplement.

Ablations and Analyses

Policy-side per-B gates and the frozen recent-dose sweep appear in Results. Table 3 brackets the learned selector between allocation rules that share its budget and rendering exactly. Random exact-B and RGB-similarity restoration are indistinguishable from Recent-B: sparse exposure of archived pixels, or visual similarity to the current screen, is not utility. Inverting the learned score is significantly harmful at both budgets—a specificity control showing the scorer ranks real signal symmetrically rather than exploiting a rendering artifact. Retraining without the recency-membership features returns the selector to the noise band, so budget-context features carry substantial weight.

The budget is spent adaptively: at B=4 the learned scorer keeps the entire recent window on 19% of states and replaces all of it on 26%—a state-dependent k profile that neither random allocation nor RGB similarity reproduces.

Closed-loop budget sweep. [Pending: Replace Table 4 with final multi-run MobileWorld results at $B \in \{1, 2, 8\}$ and paired confidence intervals. Until completion, all unfinished cells remain placeholders rather than partial point estimates.]

Limitations and Conclusion

We control an active context budget, not archive capacity: every compared policy exposes exactly B summary-plus-image events, and CausalCache reattaches rather than generates pixels. Restoration moves official retained turns—verbatim response plus screenshot—as a unit, so gains attribute to that bundle rather than to pixels alone; states whose critical value is recoverable only from pixels are rare in our corpus (0.36% at B=4), and a finer decomposition would require off-protocol probes. We claim only that some summarized events have greater conditional marginal utility than the recent realization they displace; "causal" denotes controlled prompt interventions, not structural causality, and teacher-forced margins must be judged by closed-loop success. We report a true-previous-frame control for the duplicated training-time B=1 slot; pretraining contamination remains possible. In one sentence: CausalCache learns which summarized GUI events deserve to be seen again.

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